CURRENT ELECTRICITY

⇒ Electric Current :- The rate of directional flow of electric charge.

- Scaler quantity
- Moves in the direc^n of motion of +ve charge
- Unit (Ampere → A), dimension $[A^1]$

Electric current

At t — in t

$\boxed{I_{nst} = \frac{dq}{dt}}$ slope of q/t graph

$\boxed{I_{Avg} = \frac{\Delta q}{\Delta t} = \frac{q_f - q_i}{\Delta t}}$

$\boxed{I_{Avg} = \frac{\int I dt}{\int dt}}$

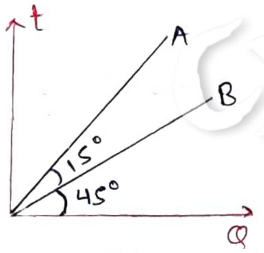
Area = नाली $\quad \Delta Q = \int I dt$ = Area of I/t graph

q (charge) ⟶ I (current)

$I = \frac{dq}{dt}$ = slope (q-t) graph $\quad \frac{dI}{dt}$ = नाली

Q⇒ Ratio of current through wire A to B

$\frac{I_A}{I_B} = \frac{\tan 30°}{\tan 45°} = \frac{1}{\sqrt{3}}$

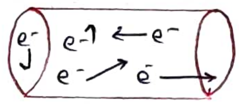


Q⇒ If 'n' electron moving on circular path with frequency 'f', find current?

$\boxed{I = nef}$

Isolated conductor

- Not connected to the battery.
- large no. of e^- are in random motion.



- $ne^- = 10^{28} e^-/m^3$

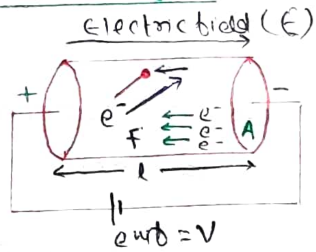
n = no. of free e^- per unit volume
└ unit = m^{-3}

- $E_{in} = 0$; $V = 10^3 - 10^4$ m/s
- $V_{avg} = 0$
- speed ∝ $\sqrt{Temp}$
- $\frac{1}{2}mv^2 = \frac{3}{2}K_BT$
- Temp ↑ → τ ↓

τ (relaxation time)
└ Avg. time b/w two collision

$K_B = 1.38 \times 10^{-23} \frac{J}{K}$

Battery connected to conductor



$F_{e^-} = qE = eE, \quad E = \frac{V}{l}$

$a = \frac{qE}{m_e} = \frac{eE}{m_e}$ (uniform)

$V_D = \cancel{u}^{0} + at$

$\boxed{V_D = \frac{eE}{m}\tau} \quad \boxed{I = neAV_d}$

Ohm's law :- $\boxed{V = IR}$

Resistivity / Resistance / Conductance

$\boxed{\sigma = \frac{ne^2\tau}{m}}$

Resistivity ⟶ $\boxed{\rho = \frac{1}{\sigma}}$ ⟶ Conducta.
↓
Material properties

$\boxed{\rho = \frac{m}{ne^2\tau}} \quad \boxed{R = \frac{\rho l}{A}}$

MR* Feel ***

$\sigma = \frac{ne^2\tau}{m} \xrightarrow{\rho = \frac{1}{\sigma}} \rho = \frac{m}{ne^2\tau} \xrightarrow{R = \frac{\rho l}{A}} R = \frac{ml}{ne^2\tau A}$

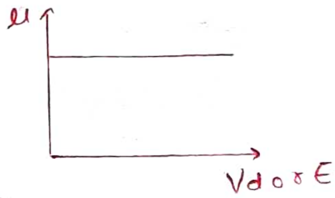
$V = \frac{mlI}{ne^2\tau A}$ ⟵ $V = IR$

Microscopic form of ohm's law — Macroscopic form of ohm's law

Mobility :- property of charge particle.

- does not depend on drift velocity & Electric field.

$\boxed{\mu = \frac{V_d}{E} = \frac{e\tau}{m}}$



- $\mu_e > \mu_p > \mu_D = \mu_\alpha$

Current density :- (J)

- vector, unit → Amp/m²
- direction along current.

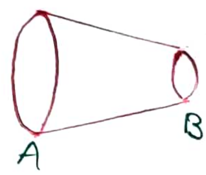
$\boxed{J = \frac{I}{A}} \quad I = \vec{J} \cdot \vec{A} = JA\cos\theta$

$\vec{J} = ne\vec{V_d} \quad \vec{J} = \sigma\vec{E}$ → vector form of ohm's law

Current :- $I_A = I_B$

current density :- $J_A < J_B$

Drift velocity :- $V_A < V_B$



★ NOTE :-

If Temp ↑

- **Conductor**
  - $\sigma \downarrow$ $\rho \uparrow$
  - $V_d \downarrow$ $\tau \downarrow$
  - Randomness ↑
- **Insulator**
  - Same as semi-conductor but in small Amount
- **Semi-conductor**
  - e^--hole pair ↑
  - $\sigma \uparrow$
  - $\rho \downarrow$

Resistance

$R \propto l \propto \frac{1}{A}$ $\quad \boxed{R = \frac{\rho l}{A}}$

① If Area(A) = constant

$R \propto l$ $\quad \boxed{\frac{R_1}{R_2} = \frac{l_1}{l_2}}$

$\left[100 \times \frac{\Delta R}{R}\right] = \left[\frac{\Delta l}{l} \times 100\right]$ → for small change

② If Length (L) = constant

$R \propto \frac{1}{A}$ $\quad \boxed{\frac{R_1}{R_2} = \frac{A_2}{A_1} = \left(\frac{r_2^2}{r_1^2}\right)}$

$\frac{\Delta R}{R} = \frac{-\Delta A}{A}$ → $\frac{\Delta R}{R} = -2\frac{\Delta r}{r}$

③ If Volume (V) = constant (stretched)

$R = \frac{\rho l}{A} \times \frac{l}{l}$

$R = \frac{\rho l^2}{V}$

$R \propto l^2$

$\boxed{\frac{R_1}{R_2} = \left(\frac{l_1}{l_2}\right)^2}$

$\frac{\Delta R}{R} = 2\frac{\Delta l}{l}$

$R = \frac{\rho(l \times A)}{A \; A} = \frac{\rho V}{A^2}$

$R \propto \frac{1}{A^2} \propto \frac{1}{r^4}$

$\frac{\Delta R}{R} = -2\frac{\Delta A}{A} = -4\frac{\Delta r}{r}$

$\boxed{\frac{R_1}{R_2} = \left(\frac{A_2}{A_1}\right)^2 = \left(\frac{r_2}{r_1}\right)^4}$

Temp^r coefficient of linear expansion

$\alpha = \frac{\Delta l}{l \Delta T}$

Temp^r coefficient of Areal expansion

$\beta = \frac{\Delta A}{A \Delta T}$

Relation b/w temp^r coefficient of

ⓐ Linear expansion

ⓑ Areal expansion

$\beta = 2\alpha$

Temp^r coefficient of Resistance

$\alpha_R = \frac{\Delta R}{R_0 T}$

R_t = Resistance at temp^r T

R_0 = Resist. at 0°C

$R_t = R_0(1 + \alpha T)$

Temp^r coefficient of Resistivity

$\boxed{\alpha_\rho = \frac{\Delta \rho}{\rho_0 \Delta T}}$

Relation b/w temp^r coefficient of:-

(a) linear expansion (α)

(b) Resistance (α_R)

(c) Resistivity (α_ρ)

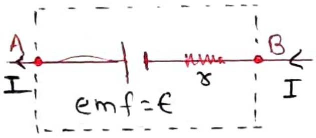
$R = \frac{\rho l}{A}$

$\alpha_R = \alpha_\rho - \alpha$

Cell (Battery)

Terminal potential in discharging of Battery :-

$V_T = (V_A - V_B) = \mathcal{E} - Ir$



Ideal Battery :- (r = 0)

$V_T = \mathcal{E} - I \times 0$ $\quad V_T = \mathcal{E}$

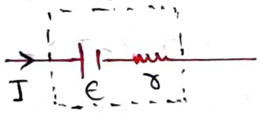
Not connected in circuit:-

$V_T = \mathcal{E} - Ir = \mathcal{E} - 0 \times r$

$V_T = \mathcal{E}$

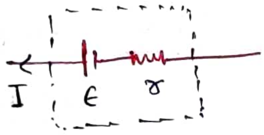
Charging of Battery:-

$\boxed{V_T = \mathcal{E} + Ir}$



Discharging of Battery

$\boxed{V_T = \mathcal{E} - Ir}$



Combination of Resistance

① In series :- [I = same], [V = diff]

$V_{total} = V_1 + V_2 + V_3 + \ldots$

$R_{eq} = R_1 + R_2 + R_3 + \ldots$

⇒ If n equal Resistance

$R_{eq} = nR$

⇒ $V \propto R$ → Jitna jyada 'R' utna jyada Vdrop.

② In parallel :- [V = same], [I = diff.]

$I_{total} = I_1 + I_2 + I_3 + \ldots$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots$$

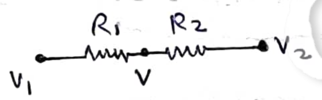
⇒ If 'n' equal resistance $R_{eq} = \frac{R}{n}$

↳ Household circuit = Parallel circuit

⇒ $I \propto \frac{1}{R}$ → Jitna kam 'R' Utna jyada 'I'.

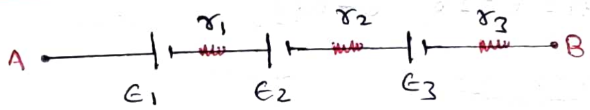
Potential at Mid-Point

$$\boxed{V_{mid} = \frac{\frac{V_1}{R_1} + \frac{V_2}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}}}$$



Combination of cell

① Series combination of cell



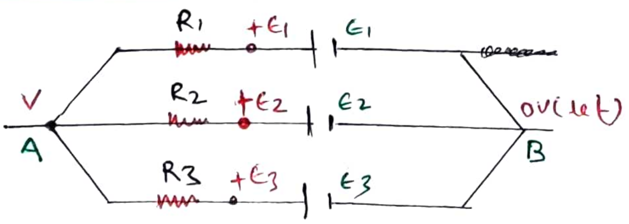
$E_{net} = E_1 + E_2 + E_3$

$r_{net} = r_1 + r_2 + r_3$

⇒ If there are 'n' identical battery is connected in series.

$$\boxed{E_{net} = nE, \quad r_{net} = nr}$$

② Parallel combination of cell



$$\boxed{E_{eq} = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}} \qquad \boxed{\frac{1}{r_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

$$\boxed{V_{AB} = (V-0) = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}}$$

⇒ If there are 'n'-identical cell in parallel

$$\boxed{E_{net} = E\ (\text{const}), \quad r_{eq} = r/n}$$

③ Mixed combination of cell

↳ n-identical battery (E, r) connected in series, then this series combination connected m-times parallel with external resistance R.

$$I = \frac{nmE}{mr + nr}$$

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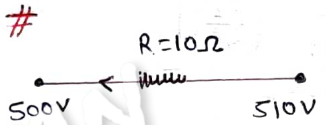
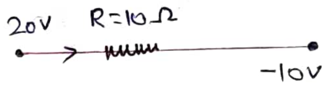




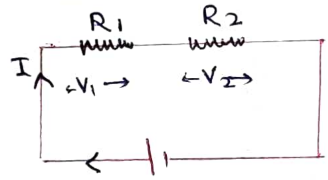
$I = \frac{510 - 500}{10}$

$I = 1A$

$I = \frac{20 - (-10)}{10} = 3A.$

⇒ Find potential drop across R_1 and R_2

$I = \frac{V}{R_1 + R_2}$



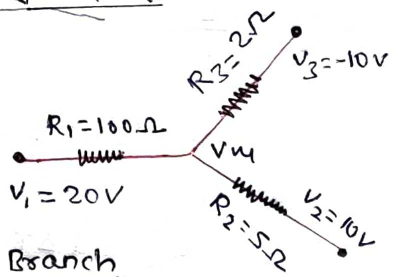
$I = \frac{V}{R_1 + R_2}$

$V_1 = IR_1 = \frac{VR_1}{R_1 + R_2}$

$V_2 = \frac{VR_2}{R_1 + R_2}$

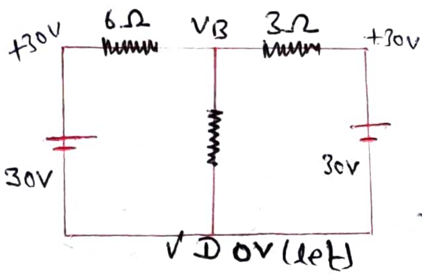
⇒ Find potential at junction

$$\boxed{V_M = \frac{\frac{20}{10} + \frac{10}{5} - \frac{10}{2}}{\frac{1}{10} + \frac{1}{5} + \frac{1}{2}}}$$

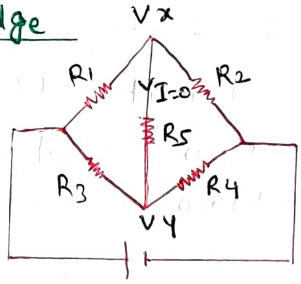


⇒ Find current in BD Branch

$$V = \frac{\frac{30}{6} + \frac{0}{3} + \frac{30}{3}}{\frac{1}{3} + \frac{1}{6} + \frac{1}{3}}$$



Wheat stone bridge



$V_x = V_y$

Current in $R_5 = 0$

$$\boxed{\frac{R_1}{R_3} = \frac{R_2}{R_4}}$$

Power

- Scaler, unit → watt (w),

$E = Pt$

$I^2Rt = Pt$

$$\boxed{P = I^2R = \frac{V^2}{R} = IV}$$

Joules law of Heating effect | Power loss

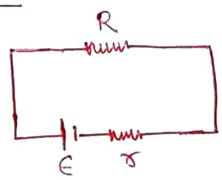
Heat (H) $\propto I^2$
$\propto R$
$\propto t$

$$\boxed{H = I^2Rt = \left(\frac{V^2}{R}\right)t = VIt}$$

$H = m\Delta\theta s$

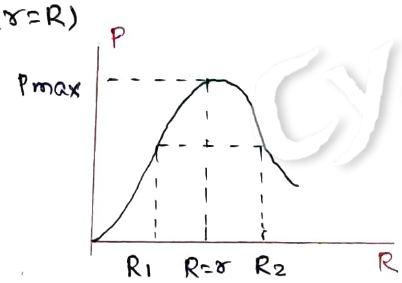
Maximum Power theorem

$$\boxed{P = \frac{\epsilon^2}{(R+r)^2}R}$$

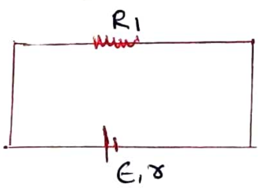


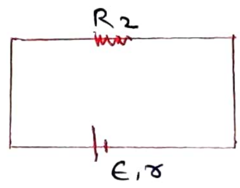
Power drop will be maximum when $(r=R)$

$$\boxed{P_{max} = \frac{\epsilon^2}{4R}}$$



⇒ If Power drop is same in both R_1 & R_2 then find Reln b/w R_1, R_2 and r.





$I_1 = \frac{\epsilon}{R_1+r}$

$I_2 = \frac{\epsilon}{R_2+r}$

$$\frac{\sqrt{R_1}}{R_1+r} = \frac{\sqrt{R_2}}{(R_2+r)} \Rightarrow \boxed{r = \sqrt{R_1R_2}}$$

Time taken to boil same amount of water when connected in

ⓐ Series

$$\boxed{t = t_1 + t_2}$$

ⓑ Parallel

$$\boxed{\frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2}}$$

Bulb :- (Pure resistance hai)

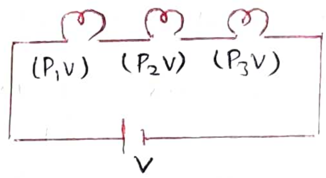
$R_{bulb} = \frac{V_R^2}{P_R}$

$$\boxed{P_{cons.} = \left[\frac{V_S}{V_R}\right]^2 P_{Rated}}$$

$R_{bulb} \propto \frac{1}{P_{rated}}$

① Series combination

$$\boxed{\frac{1}{P_{cons.}} = \frac{1}{P_1} + \frac{1}{P_2} + \frac{1}{P_3}}$$



If All are identical, bulb

$$\boxed{P_{cons.} = \frac{P}{n}}$$

$$\boxed{\text{Brightness} \propto P_{cons.} \propto R_{bulb} \propto \frac{1}{P_{rated}}}$$

$$\boxed{P_{cons.} = i^2 R_{bulb}}$$

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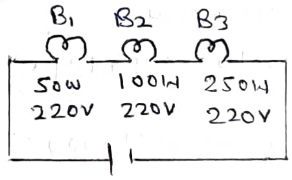
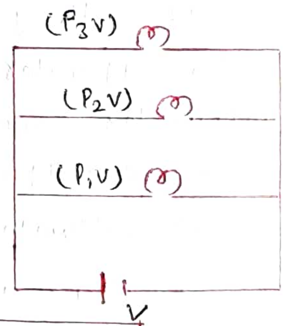


Brightness
$(B_3 > B_1 > B_2)$

② Parallel combination

$P_{cons.} = P_1 + P_2 + P_3$



If all are identical bulb

$$\boxed{P_{cons} = nP}$$

$$\boxed{\text{Brightness} \propto P_{cons} \propto \frac{1}{R_{bulb}} \propto P_{rated}}$$

$$\boxed{P_{consumed} = \frac{V^2}{R_{bulb}}}$$

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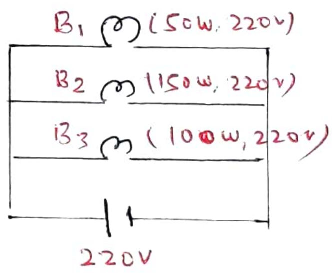
Brightness
$(B_2 > B_3 < B_1)$



Galvanometer

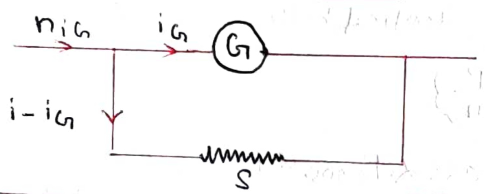
- An instrument use to detect or measure small current
- Very sensitive, produce large errors

I_G = Max^m current that can flow
G = resistance of galvanometer.

⇒ **As Ammeter :-** (Measure current)

- Shunt (small) connect in parallel with galvanometer.
- Ammeter connected in series with circuit to measure current.
- Ideal $R = 0$; Behave as simple wire

$$\boxed{\%\ \text{Error} = \frac{i_T - i_M}{i_T} \times 100}$$



$$\boxed{R_{ammeter} = \frac{GS}{G+S}} \qquad S = \frac{G}{n-1}$$

$$\boxed{n = \frac{(I)\ \text{we want to Measure}}{(I)\ \text{jitna Galvanometer se Jayega}}}$$

⇒ **As Voltmeter :-** (Measure voltage)

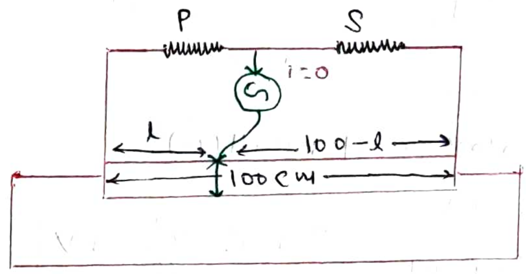
- Shunt (large) resistance connected in series with galvanometer to convert into voltmeter
- Voltmeter connected in parallel to measure voltege.
- Ideal voltmeter (R = Infinite)
  ↳ Behave as open wire

$$\boxed{\%\ \text{Error} = \frac{V_T - V_M}{V_T} \times 100}$$

$$\boxed{R_{voltmeter} = G + S} \qquad S = G(n-1)$$

Meter Bridge

- An instrument used to measure resistance
- Based on concept of wheat stone bridge.
- Resistance per unit length $\lambda = \left(\frac{R}{L}\right)$



$$\frac{P}{Q} = \frac{l}{100-l}$$